UNIVERSITÀ di VERONA
Dipartimento di INFORMATICA

Discrete Optimization and Decision Making

Modeling in Linear Programming

Master's degrees in Data Science - Artificial Intelligence - Mathematics
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Romeo Rizzi (Dept. of Computer Science, Univ. of Verona)
Roberto Zanotti (Dept. of Clinical and Experimental Sciences, Univ. of Brescia)

The transportation problem

m plants;

n customers;

s_i = production (capacity) of plant i (# of units);

d_j = request (demand) of customer j (# of units);

c_{ij} = unitary cost of transportation from i to j (directly proportional to the transported quantity).

DECISION VARIABLES:

$$\begin{matrix} x_{11} & x_{12} & x_{13} & \cdots & x_{1n} \\ x_{21} & x_{22} & x_{23} & \cdots & x_{2n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ x_{m1} & x_{m2} & x_{m3} & \cdots & x_{mn} \end{matrix}$$

x_{ij}= transported quantity from i to j.

OBJECTIVE: **minimize the total transportation costs**.

$$\downarrow$$

$$min \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij} x_{ij}$$

PRODUCTION CONSTRAINTS:

$$\sum_{j=1}^{n} x_{ij} \leq s_i \qquad i = 1, \cdots, m \quad (\rightarrow m \text{ constraints})$$

$$x_{11} + x_{12} + x_{13} + \cdots + x_{1n} \leq s_1$$

$$x_{21} + x_{22} + x_{23} + \cdots + x_{2n} \leq s_2$$

$$\vdots$$

$$x_{m1} + x_{m2} + x_{m3} + \cdots + x_{mn} \leq s_m$$

DEMAND CONSTRAINTS:

$$\sum_{i=1}^{m} x_{ij} \geq d_j \qquad j = 1, \cdots, n \quad (\rightarrow n \text{ constraints})$$

$$x_{11} + x_{21} + x_{31} + \cdots + x_{m1} \geq d_1$$

$$x_{12} + x_{22} + x_{32} + \cdots + x_{m2} \geq d_2$$

$$\vdots$$

$$x_{1n} + x_{2n} + x_{3n} + \cdots + x_{mn} \geq d_n$$

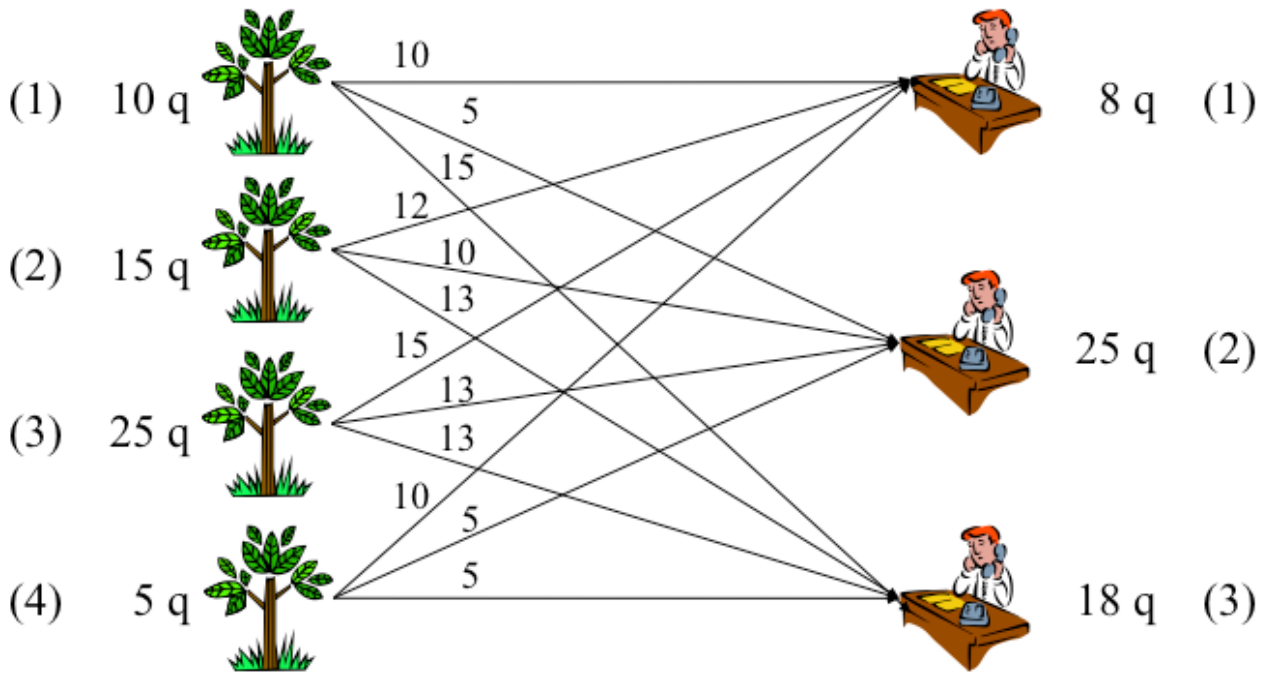
NON-NEGATIVITY CONSTRAINTS:

$$x_{ij} \geq 0 \qquad i = 1, \cdots, m \quad j = 1, \cdots, n$$

NECESSARY CONDITION TO ENSURE FEASIBILITY:

$$\sum_{i=1}^{m} s_i \geq \sum_{j=1}^{n} d_j$$

A numerical example



$$x_{11} + x_{12} + x_{13} \leq 10$$

$$x_{21} + x_{22} + x_{23} \leq 15$$

$$x_{31} + x_{32} + x_{33} \leq 25$$

$$x_{41} + x_{42} + x_{43} \leq 5$$

$$x_{11} + x_{21} + x_{31} + x_{41} \geq 8$$

$$x_{12} + x_{22} + x_{32} + x_{42} \geq 25$$

$$x_{13} + x_{23} + x_{33} + x_{43} \geq 18$$

$$x_{11}, x_{12}, x_{13}, x_{21}, x_{22}, x_{23}, x_{31}, x_{32}, x_{33}, x_{41}, x_{42}, x_{43} \geq 0$$

Let's make a change to the demand constraints:

$$-x_{11} - x_{21} - x_{31} - x_{41} \leq -8$$

$$-x_{12} - x_{22} - x_{32} - x_{42} \leq -25$$

$$-x_{13} - x_{23} - x_{33} - x_{43} \leq -18$$

The associated **constraints matrix** then is:

1	1	1	0	0	0	0	0	0	0	0	0
0	0	0	1	1	1	0	0	0	0	0	0
0	0	0	0	0	0	1	1	1	0	0	0
0	0	0	0	0	0	0	0	0	1	1	1
-1	0	0	-1	0	0	-1	0	0	-1	0	0
0	-1	0	0	-1	0	0	-1	0	0	-1	0
0	0	-1	0	0	-1	0	0	-1	0	0	-1

Observation:

In each column there are at most 2 elements different from 0 (specifically a 1 and a -1).

$$\downarrow$$

A matrix with this peculiar structure (with some additional conditions), is called **totally unimodular**, a feature that makes this problem *easy* to solve, despite the need for the variables to be integer.

The production mix problem

A company sells three different dried fruits assortments, each one with a different quantity of almonds, peanuts, walnuts, and cashews.

Assortments	Composition requirements	Sale prices per pound
Regular	At most 20% of cashews At least 40% of walnuts At most 25% of peanuts	59
Deluxe	At most 35% of cashews At least 25% of almonds	69
Blue Ribbon	Between 30% and 50% of cashews At least 30% of almonds	85

Ingredients costs:

Type	Cost per pound	Maximum available quantity per week (in pounds)
Almonds	25	2000
Peanuts	35	4000
Cashews	50	5000
Walnuts	30	3000

PROBLEM: determine the exact amount of each dried fruit to put in each assortment, in order to maximize profits.

VARIABLES: quantity of each type of dried fruit in the composition;

CONSTRAINTS: maximum quantity of each ingredient in the composition and availability of the resources;

$$\Downarrow$$

OBJECTIVE: **maximize the company's profits**.

DECISION VARIABLES:

r_i = quantity in the Regular mix for the dried fruit type $i = 1, ..., 4$;
d_i = quantity in the Deluxe mix for the dried fruit type $i = 1, ..., 4$;
b_i = quantity in the Blue Ribbon mix for the dried fruit type $i = 1, ..., 4$.

OBJECTIVE FUNCTION: *max* profit = *max* (tot. revenues - tot. costs)

Total revenues
$= 59(r_1 + r_2 + r_3 + r_4) + 69(d_1 + d_2 + d_3 + d_4) + 85(b_1 + b_2 + b_3 + b_4)$

Total costs
$= 25(r_1 + b_1 + d_1) + 35(r_2 + b_2 + d_2) + 50(r_3 + b_3 + d_3) + 30(r_4 + b_4 + d_4)$

Profitt = $T_r - T_C$ =
$34r_1 + 24r_2 + 9r_3 + 29r_4 + 44d_1 + 34d_2 + 19d_3 + 39d_4 + 60b_1 + 50b_2 + 35b_3 + 55b_4$

CONSTRAINTS on the maximum quantities:

$$r_1 + d_1 + b_1 \leq 2000$$

$$r_2 + d_2 + b_2 \leq 4000$$

$$r_3 + d_3 + b_3 \leq 5000$$

$$r_4 + d_4 + b_4 \leq 3000$$

CONSTRAINTS on the assortments percentages:

$$r_3 \leq 0.2(r_1 + r_2 + r_3 + r_4)$$
$$r_4 \geq 0.4(r_1 + r_2 + r_3 + r_4)$$
$$r_2 \leq 0.25(r_1 + r_2 + r_3 + r_4)$$

$$d_3 \leq 0.35(d_1 + d_2 + d_3 + d_4)$$
$$d_1 \geq 0.25(d_1 + d_2 + d_3 + d_4)$$

$$b_3 \geq 0.3(b_1 + b_2 + b_3 + b_4)$$
$$b_3 \leq 0.5(b_1 + b_2 + b_3 + b_4)$$

$$b_1 \geq 0.3(b_1 + b_2 + b_3 + b_4)$$

$$r_i, d_i, b_i \geq 0, \quad i = 1, ..., 4$$

The Diet Problem

m nutrients;

n types of food;

a_{ij}= quantity of nutrient i in food j

b_i= quantity of nutrient i required by the diet

c_j= unit cost of food j

SUFFICIENT CONDITION (for the existence of a feasible diet): each nutrient has to be included in at least one food. In this case, an optimal diet also exists.

DECISION VARIABLES:
x_j= quantity of food j in the diet

OBJECTIVE FUNCTION:

$$min \sum_{j=1}^{n} c_j x_j$$

CONSTRAINTS:

$$\sum_{j=1}^{n} a_{ij} x_j \geq b_i \qquad i = 1, \cdots, m$$

in detail:

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \geq b_1$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \geq b_m$$

$$x_j \geq 0 \qquad j = 1, \cdots, n$$

A numerical example

Model the following diet problem, based on the data available in the following table:

	Fillet	Pecorino	Eggplant	Minimum Required
Protein	50	15	3	26
Vitamin A	10	8	9	80
Vitamin B	6	5	12	30
Cost (per Kg)	15	8	3	

DECISION VARIABLES:

x_f= quantity of fillet to buy;

x_p= quantity of pecorino to buy;

x_m= quantity of eggplant to buy;

MODEL:

$$\min 15x_f + 8x_p + 3x_m$$

$$50x_f + 15x_p + 3x_m \geq 26$$

$$10x_f + 8x_p + 9x_m \geq 80$$

$$6x_f + 5x_p + 12x_m \geq 30$$

$$x_f \geq 0, x_p \geq 0, x_m \geq 0$$

An Investment problem

At the beginning of month 1, the company Felp S.p.A. has 400$ in cash. At the beginning of each month, the company receives a certain amount of income that it uses to pay for expenses incurred.

	INCOME $	EXPENSES $
month 1	400	600
month 2	800	500
month 3	300	500
month 4	300	250

All the money left over from paying expenses can be invested for one month at an interest rate of 0.1% per month, for two months at a rate of 0.5% per month, for three months at a rate of 1% per month, or for four months at an interest rate of 2% per month. Formulate a linear programming model that maximizes the amount available as cash at the beginning of month 5.

Decision variables:

$$x_{ij} = \text{investment in month } i \text{ for } j \text{ months}$$

Objective function:

$$\max\ 1.001x_{41} + (1.005)^2 x_{32} + (1.01)^3 x_{23} + (1.02)^4 x_{14}$$

Constraints:

$$\begin{aligned}
x_{11} + x_{12} + x_{13} + x_{14} + 600 &= 400 + 400 \\
x_{21} + x_{22} + x_{23} + 500 &= 800 + 1.001x_{11} \\
x_{31} + x_{32} + 500 &= 300 + 1.001x_{21} + (1.005)^2 x_{12} \\
x_{41} + 250 &= 300 + 1.001x_{31} + (1.005)^2 x_{22} + (1.01)^3 x_{13} \\
x_{ij} &\geq 0\ \forall i, j
\end{aligned}$$

NB: Interest rate at the $n-$month = $(1 + r)^n$, r monthly interest rate.

Linearization of function `MIN`

Given a set of continuous variables $x_1, x_2, ..., x_m \in \mathbb{R}$, we want to find the one with minimum value, and assign it to a new variable w:

$$w = \min\{x_1, x_2, ..., x_m\}$$

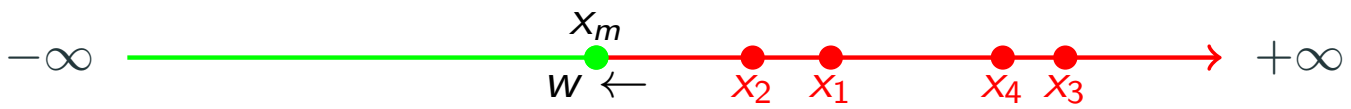
Which **linear** constraints are necessary in order for variable w to take the desired value?

Linearization of function MIN

Variable w has to be **lower** than or **equal** to every other variable in the set:

$$w \leq x_i, \quad i = 1, \ldots, m$$

In this way, we have restricted the domain of variable w:



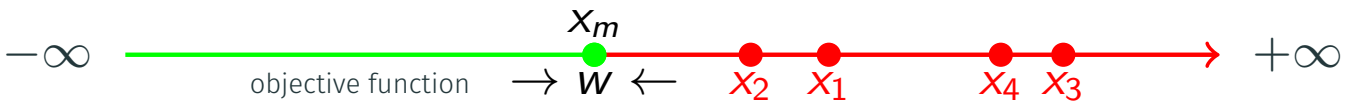
How can we assure that variable w takes exactly the minimum value, and it is not free to take any value in $(-\infty, \min\{x_1, x_2, \ldots, x_m\}]$?

Linearization of function MIN

Let's introduce a maximization function:

$$\max \quad w$$

$$w \leq x_i \quad i = 1, \ldots, m$$



Variable w is superiorly limited by the value of the variables and the objective function will force that **the value of** w **is** exactly **the maximum of the set** $(-\infty, \min\{x_1, x_2, \ldots, x_m\}]$, i.e., $\min\{x_1, x_2, \ldots, x_m\}$.

Linearization of function MAX

The approach is symmetrical:

$$w = max\{x_1, x_2, ..., x_m\}$$

$$\downarrow$$

$$min \quad w$$

$$w \geq x_i \quad i = 1...m$$

Alternatively, it could be useful to remember that

$$w = max\{x_1, x_2, ..., x_m\} = -min\{-x_1, -x_2, ..., -x_m\}$$

then...

Generalization

By generalizing the approach, it is trivial to create linear constraints for the following cases:

$$w = \min\{f_1(\mathbf{x}), f_2(\mathbf{x}), ..., f_m(\mathbf{x})\}$$

$$w = \max\{f_1(\mathbf{x}), f_2(\mathbf{x}), ..., f_m(\mathbf{x})\}$$

in which $\mathbf{x} \in \mathbb{R}^n$ and $f_i(\mathbf{x}), \quad i = 1, ..., m$ are **generic linear functions**.

Linearization of function `ABSOLUTE VALUE`

Given a continuous variable $x \in \mathbb{R}$, we want to define, through **linear** constraints, a new variable v that takes a value equal to the absolute value of x.

$$v = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases} \quad = \max\{x, -x\}$$

The linearization of this function is equivalent to the linearization of a *MAX* function!

Linearization of functions `MIN-MAX` and `MAX-MIN`

We can look at the proposed linearizations as Linear Programming models with objective function of type MIN-MAX/MAX-MIN.

$$\begin{aligned} \min \quad & w \\ w \geq x_i \quad & i = 1, \ldots, n \end{aligned} \quad \leftrightarrow \quad \begin{aligned} \min \quad & \max\{x_1, x_2, \ldots, x_n\} \\ = \ & \textbf{minimize} \text{ the } \textbf{maximum} \text{ value of } \{x_1, x_2, \ldots, x_n\} \end{aligned}$$

$$\begin{aligned} \max \quad & w \\ w \leq x_i \quad & i = 1, \ldots, n \end{aligned} \quad \leftrightarrow \quad \begin{aligned} \max \quad & \min\{x_1, x_2, \ldots, x_n\} \\ = \ & \textbf{maximize} \text{ the } \textbf{minimum} \text{ value of } \{x_1, x_2, \ldots, x_n\} \end{aligned}$$

Linearization of functions `MIN-MAX` and `MAX-MIN`

These kind of objective functions are quite common:

$$\max \quad \min\{f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_n(\mathbf{x})\}$$

$$A\mathbf{x} \geq b$$

$$\mathbf{x} \geq 0$$

The objective function has to be linearized:

- introducing and auxiliary variable $w = \min\{f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_n(\mathbf{x})\}$ (making the objective function linear)
- adding the linear constraints that force variable w to take the minimum value in the set $\{f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_n(\mathbf{x})\}$.

The case $\min - \max$ is similarly handled.

Portfolio selection problem with MAD risk measure

We want to fully invest our capital into the stock market. To determine how to invest it, we only have historical data (i.e., how the stocks performed in the past). We also want to include into our model some form of risk management, and to do that we introduce appropriate risk measures.

Mean Absolute Deviation (MAD): given the average return of an investment portfolio calculated on available historical data, MAD is the average of the absolute values of the deviations of the portfolio return in each time period from this average return.

We consider:

- $J = \{1, \dots, m\}$ set of stocks
- $T = \{1, \dots, n\}$ set of time periods
- r_{tj} return of stock j in time period t

Portfolio selection problem with MAD risk measure

Variables:

- x_j: continuous non-negative variable representing the percentage of capital to invest in stock j;
- v_t: continuous non-negative variable indicating the absolute deviation from the average return for time period t.

$$max \frac{1}{|T|} \sum_{t \in T} \sum_{j \in J} r_{tj} x_j$$

$$\sum_{j \in J} x_j = 1$$

$$v_t \geq \frac{1}{|T|} \sum_{h \in T} \sum_{j \in J} r_{hj} x_j - \sum_{j \in J} r_{tj} x_j \qquad \forall t \in T$$

$$v_t \geq \sum_{j \in J} r_{tj} x_j - \frac{1}{|T|} \sum_{h \in T} \sum_{j \in J} r_{hj} x_j \qquad \forall t \in T$$

$$\frac{1}{|T|} \sum_{t \in T} v_t \leq v^*$$

$$x_j \geq 0 \quad \forall j \in J \qquad v_t \geq 0 \quad \forall t \in T$$

What if we do the opposite?

$$min \frac{1}{|T|} \sum_{t \in T} v_t$$

$$\sum_{j \in J} x_j = 1$$

$$v_t \geq \frac{1}{|T|} \sum_{h \in T} \sum_{j \in J} r_{hj} x_j - \sum_{j \in J} r_{tj} x_j \qquad \forall t \in T$$

$$v_t \geq \sum_{j \in J} r_{tj} x_j - \frac{1}{|T|} \sum_{h \in T} \sum_{j \in J} r_{hj} x_j \qquad \forall t \in T$$

$$\frac{1}{|T|} \sum_{t \in T} \sum_{j \in J} r_{tj} x_j \geq r^*$$

$$x_j \geq 0 \quad \forall j \in J \qquad v_t \geq 0 \quad \forall t \in T$$